

Homework Answers

① $y = 3\cos(2x + 360^\circ) - 2$

Midline Equation: $y = -2$

Phase Shift: -180°

Period: 180°

Amplitude: 3

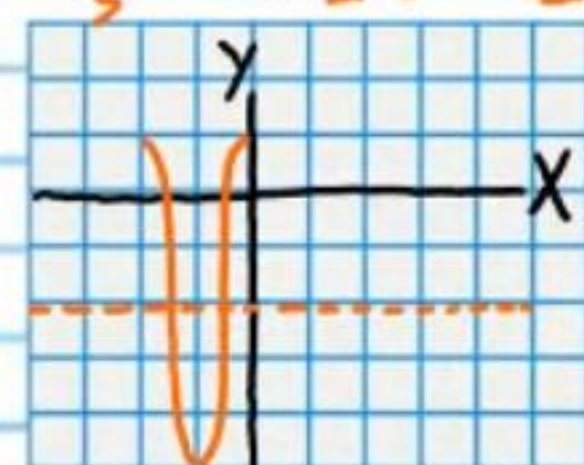
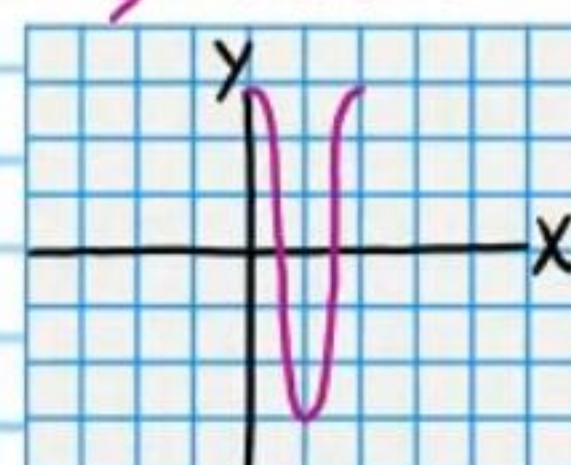
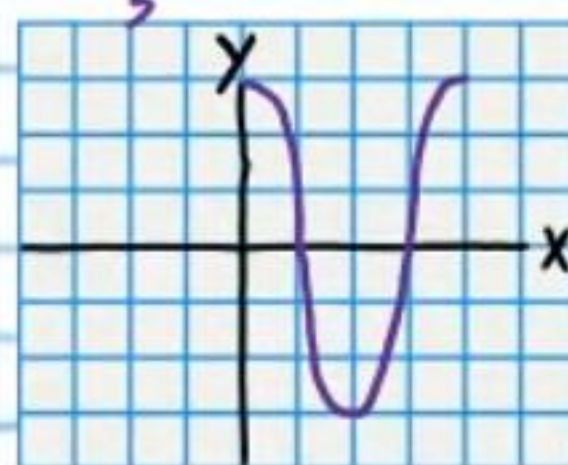
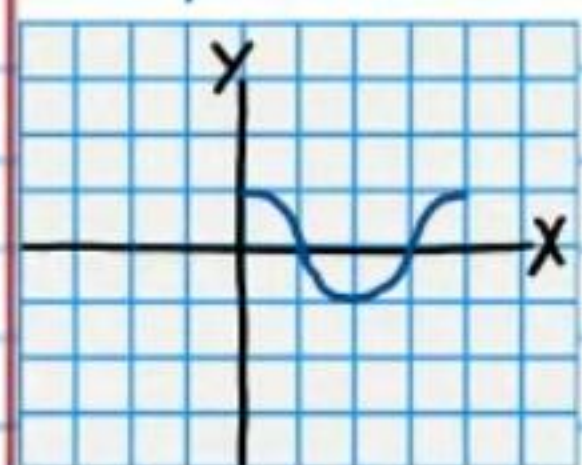
Change the form of the equation: $\frac{y+2}{3} = \cos[2(x+180^\circ)]$

Parent Function
 $y = \cos x$

Transformation #1
 $\frac{y}{3} = \cos x$

Transformation #2
 $\frac{y}{3} = \cos(2x)$

Final Transformation
 $\frac{y+2}{3} = \cos[2(x+180^\circ)]$



② $y = -2\sin(3x - 270^\circ) + 3$

Midline Equation: $y = 3$

Phase Shift: 90°

Period: 120°

Amplitude: 2

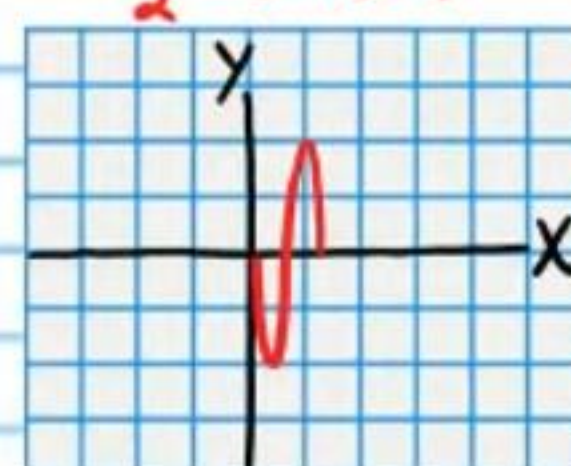
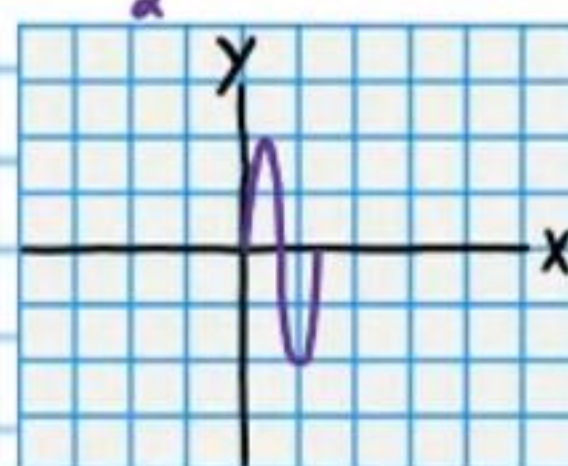
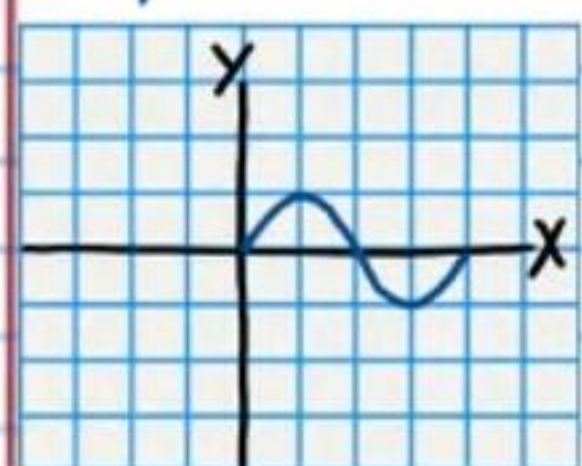
Change the form of the equation: $\frac{-(y-3)}{2} = \sin[3(x-90^\circ)]$

Parent Function
 $y = \sin x$

Transformation #1
 $\frac{y}{2} = \sin(3x)$

Transformation #2
 $-\frac{y}{2} = \sin(3x)$

Final Transformation
 $\frac{-(y-3)}{2} = \sin[3(x-90^\circ)]$



③ $y = -\cos(\frac{x}{2} - 45^\circ) - 1$

Midline Equation: $y = -1$

Phase Shift: 90°

Period: 720°

Amplitude: 1

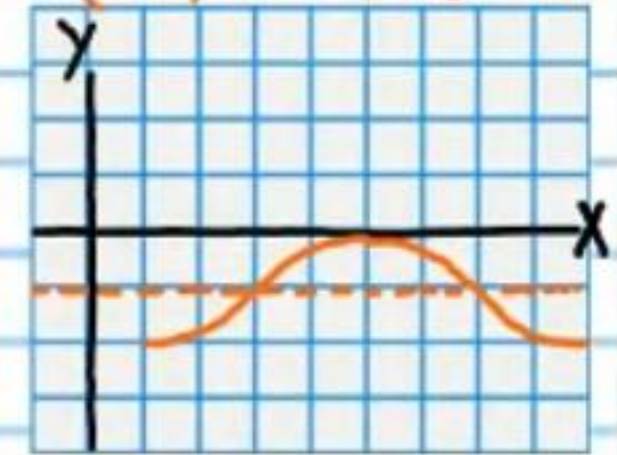
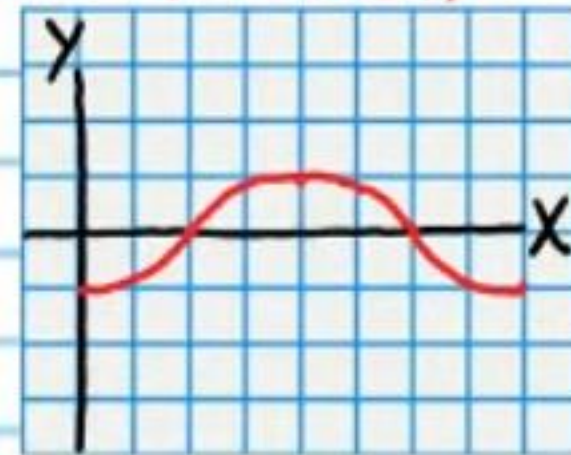
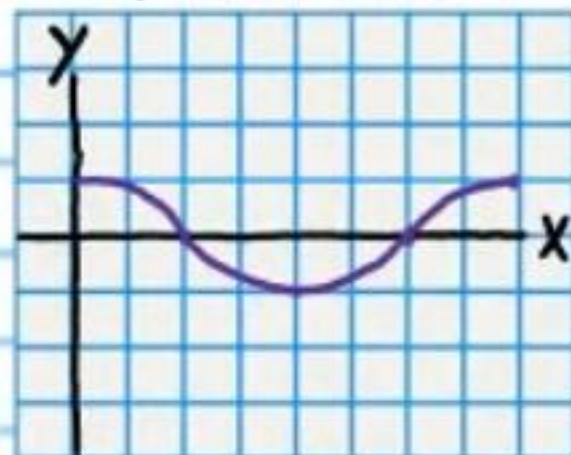
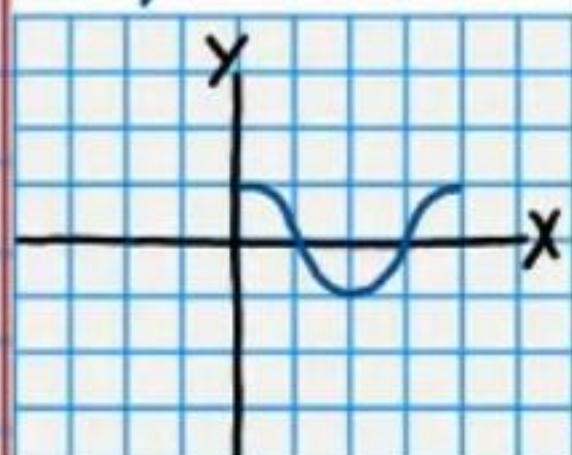
Change the form of the equation: $-(y+1) = \cos[\frac{1}{2}(x-90^\circ)]$

Parent Function
 $y = \cos x$

Transformation #1
 $y = \cos(\frac{x}{2})$

Transformation #2
 $-y = \cos(\frac{x}{2})$

Final Transformation
 $-(y+1) = \cos[\frac{1}{2}(x-90^\circ)]$



④ $y = 4\csc(\frac{x}{3} + 45^\circ) + 1$

Midline Equation: $y = 1$

Period: $1,080^\circ$

Phase Shift: -135°

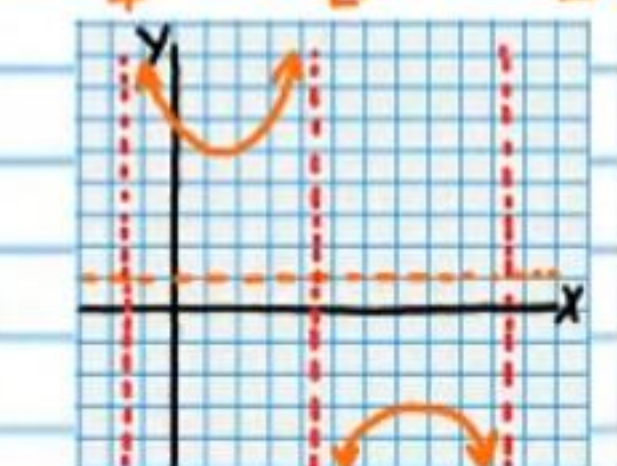
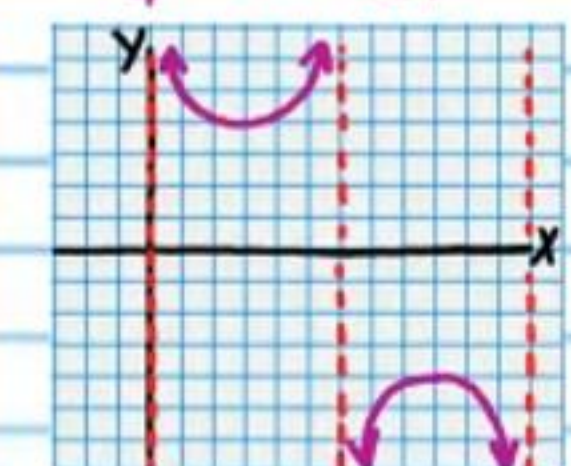
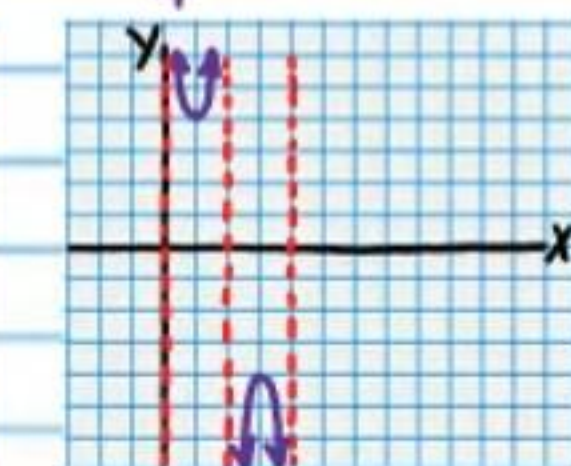
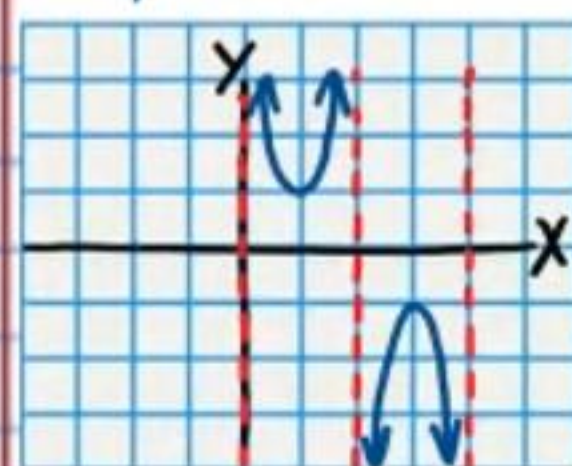
Change the form of the equation: $\frac{y-1}{4} = \csc[\frac{1}{3}(x+135^\circ)]$

Parent Function
 $y = \csc x$

Transformation #1
 $\frac{y}{4} = \csc x$

Transformation #2
 $\frac{y}{4} = \csc(\frac{x}{3})$

Final Transformation
 $\frac{y-1}{4} = \csc[\frac{1}{3}(x+135^\circ)]$



⑤ $y = -3\sec\left(\frac{x}{2} - 135^\circ\right)$

Midline Equation: $y = 0$

Phase Shift: 270°

Period: 720°

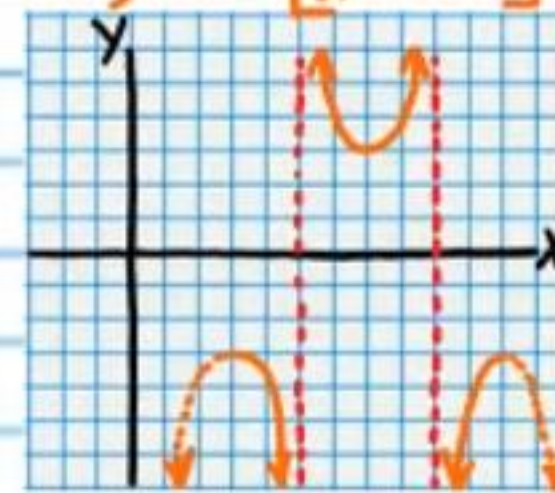
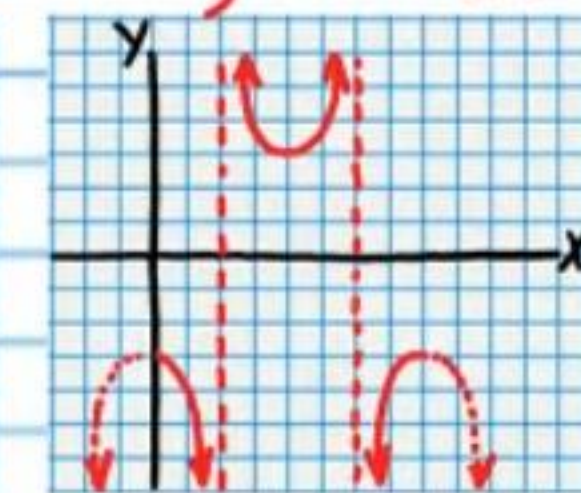
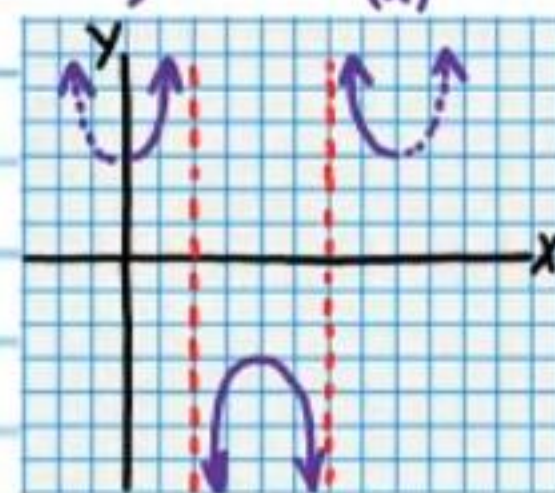
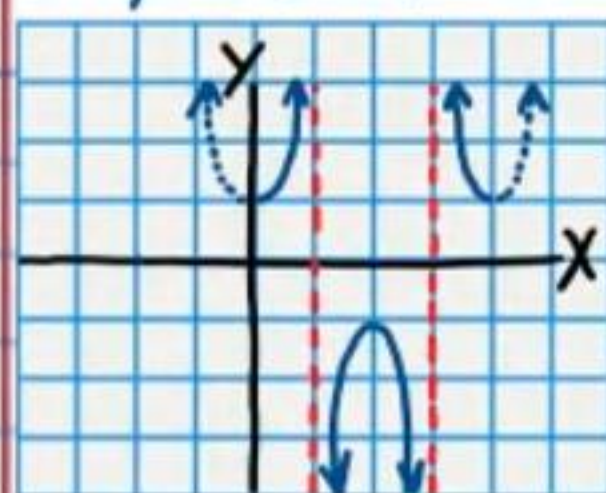
Change the form of the equation: $-\frac{y}{3} = \sec\left[\frac{1}{2}(x - 270^\circ)\right]$

Parent Function
 $y = \sec x$

Transformation #1
 $\frac{y}{3} = \sec\left(\frac{x}{2}\right)$

Transformation #2
 $-\frac{y}{3} = \sec\left(\frac{x}{2}\right)$

Final Transformation
 $-\frac{y}{3} = \sec\left[\frac{1}{2}(x - 270^\circ)\right]$



⑥ $y = 2\tan(x + 180^\circ) - 3$

Midline Equation: $y = -3$

Period: 180°

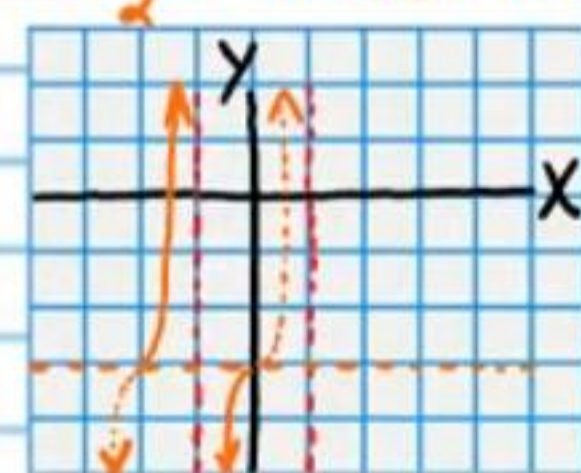
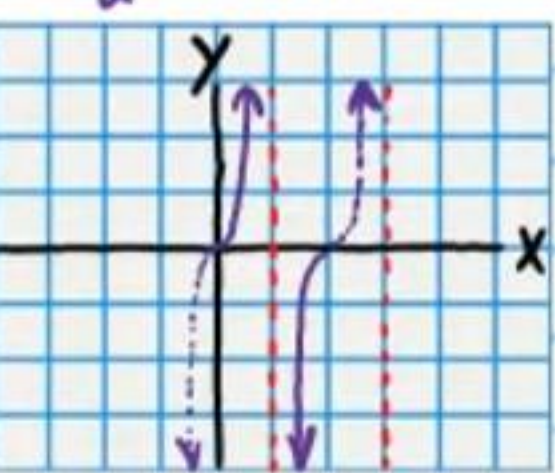
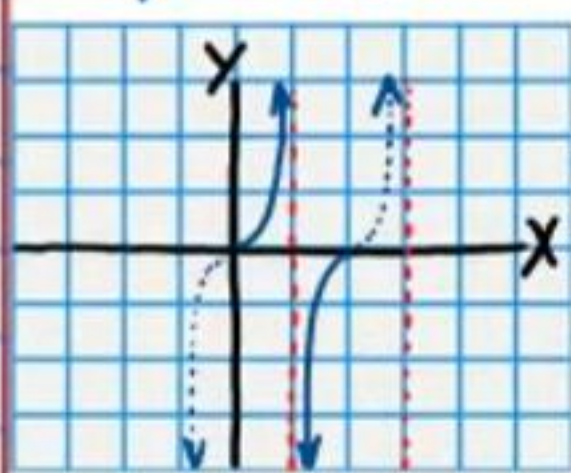
Phase Shift: -180°

Change the form of the equation: $\frac{y+3}{2} = \tan(x + 180^\circ)$

Parent Function
 $y = \tan x$

Transformation #1
 $\frac{y}{2} = \tan x$

Final Transformation
 $\frac{y+3}{2} = \tan(x + 180^\circ)$



⑦ $y = \cot\left(\frac{x}{3} - 90^\circ\right)$

Midline Equation: $y = 0$

Period: 540°

Phase Shift: 270°

Change the form of the equation: $y = \cot\left[\frac{1}{3}(x - 270^\circ)\right]$

Parent Function
 $y = \cot x$

Transformation #1
 $y = \cot\left(\frac{x}{3}\right)$

Final Transformation
 $y = \cot\left[\frac{1}{3}(x - 270^\circ)\right]$

